

What's New

Software | nand2tetris

NAND2Tetris

Ask Gemini

← → ↺

nand2tetris.github.io/web-ide/chip/

Install ☆

Begin browsing instantly: Chrome can now launch when Windows starts.

Allow

×

NAND2Tetris / Hardware Simulator / Not.hdl

HDL ☐ --LBSt

Project 1

Not

Chip Not

Eval Reset Clock: 0

```
5 /**
8 */
9 CHIP Not {
10     IN in;
11     OUT out;
12
13     PARTS:
14     //// Replace this comment with your code.
15     Nand(a=in , b=in , out=out );
16
17 }
```

Input pins

in 1

Output pins

out 0

Test Steps: 6 Outputs: 3 Slow Fast

Test Script

Compare File

Output File

Diff Table

```
12 output;
13
14 set in 1,
15 eval,
16 output;
```

Simulation successful: The output file is identical to the compare file

30°C
Sunny

Search

ENG
IN

96%

09:43 AM
23-09-26

nand2tetris.github.io/web-ide/chip/

Begin browsing instantly: Chrome can now launch when Windows starts.

Allow

NAND2Tetris / Hardware Simulator / And.hdl

HDL

--LBSt

Project 1

And

Chip And

Eval Reset Clock: 0

8 */

9 CHIP And {

10 IN a, b;

11 OUT out;

12

13 PARTS:

14 //// Replace this comment with your code.

15 Nand(a=a, b=b ,out=nandOut);

16 Not(in=nandOut , out=out);

17

18 }

Input pins

a 1

b 1

Output pins

out 1

Internal pins

nandOut 0

Test

Steps: 7 Outputs: 5

Slow

Fast

Test Script

Compare File

Output File

Diff Table

24

25 set a 1,

26 set b 1,

27 eval,

28 output;

Simulation successful: The output file is identical to the compare file

nand2tetris.github.io/web-ide/chip/

Install

Begin browsing instantly; Chrome can now launch when Windows starts.

Allow

NAND2Tetris / Hardware Simulator / Or.hdl

HDL

--LBSt

Project 1

Or

Chip Or

Eval Reset Clock: 0

8 */

9 CHIP Or {

10 IN a, b;

11 OUT out;

12

13 PARTS:

14 //// Replace this comment with your code.

15 Not(in=a, out=notA);

16 Not(in=b, out=notB);

17 Nand(a=notA, b=notB, out=out);

18 }

Input pins

a 1

b 1

Output pins

out 1

Internal pins

notA 0

Test

Steps: 7

Outputs: 5

Slow

Fast

Test Script

Compare File

Output File

Diff Table

24

25 set a 1,

26 set b 1,

27 eval,

28 output;

Simulation successful: The output file is identical to the compare file

nand2tetris.github.io/web-ide/chip/

Install

S

Begin browsing instantly: Chrome can now launch when Windows starts.

Allow

NAND2Tetris / Hardware Simulator / Xor.hdl

<>

HDL

--LBSt

Project 1

Xor

Chip Xor

Eval

Reset

Clock: 0

9

CHIP Xor {

10

IN a, b;

11

OUT out;

12

13

PARTS:

14

//// Replace this comment with your code.

15

Or(a=a, b=b, out=orAB);

16

And(a=a, b=b, out=andAB);

17

Not(in=andAB, out=notAndAB);

18

And(a=orAB, b=notAndAB, out=out);

19

}

Input pins

a

1

b

1

Output pins

out

0

Internal pins

orAB

1

Test

Steps: 7

Outputs: 5

Slow

Fast

Test Script

Run

Compare File

Output File

Diff Table

24

25 set a 1,

26 set b 1,

27 eval,

28 output;

Simulation successful: The output file is identical to the compare file

Eval Reset Clock: 0

Internal pins

Fast

Diff Table

Simulation successful: The output file is identical to the compare file

Eval Reset Clock: 0

Internal pins

 Fast

Diff Table

```
26 output;
```

Simulation successful: The output file is identical to the compare file

Test Steps: 8 Outputs: 6

Slow Fast

Test Script | Compare File | Output File | Diff Table

```

1 00000000:
2 7 compare-to Not16.cmp,
3 8 output-list in%81.16.1 out%81.16.1;
4 9
5 10 set in %800000000000000000,
6 11 eval,
7 12 output;
8 13
9 14 set in %811111111111111111,
10 15 eval,
11 16 output;
12 17
13 18 set in %81010101010101010,
14 19 eval,
15 20 output;
16 21
17 22 set in %80011110011000011,
18 23 eval,
19 24 output;
20 25
21 26 set in %80001001000110100,
22 27 eval,
23 28 output;

```

30°C Sunny Search [Taskbar icons: File Explorer, Edge, Word, PowerPoint, Outlook, Teams, OneDrive, Chrome] ENG IN 90% 09:56 AM 23-09-26

Software | nand2tetris

NAND2Tetris

ALU Code Request

sg562656-sudo/Nand2Tetris

Ask Gemini

nand2tetris.github.io/web-ide/chip/

Begin browsing instantly: Chrome can now launch when Windows starts. Allow

NAND2Tetris / Hardware Simulator / And16.hdl

Chip And16 Eval Reset Clock: 0

Test Steps: 9 Outputs: 7

HDL

--LBSt

Project 1

And16

5 /**

10 CHIP And16 {

11 IN a[16], b[16];

12 OUT out[16];

13

14 PARTS:

15 // Replace this comment with your code.

16 And(a=a[0], b=b[0], out=out[0]);

17 And(a=a[1], b=b[1], out=out[1]);

18 And(a=a[2], b=b[2], out=out[2]);

19 And(a=a[3], b=b[3], out=out[3]);

20 And(a=a[4], b=b[4], out=out[4]);

21 And(a=a[5], b=b[5], out=out[5]);

22 And(a=a[6], b=b[6], out=out[6]);

23 And(a=a[7], b=b[7], out=out[7]);

24 And(a=a[8], b=b[8], out=out[8]);

25 And(a=a[9], b=b[9], out=out[9]);

26 And(a=a[10], b=b[10], out=out[10]);

27 And(a=a[11], b=b[11], out=out[11]);

28 And(a=a[12], b=b[12], out=out[12]);

29 And(a=a[13], b=b[13], out=out[13]);

30 And(a=a[14], b=b[14], out=out[14]);

31 And(a=a[15], b=b[15], out=out[15]);

32

33 }

Input pins

a 0 0 0 1 0 0 1 0 0 0 1 1 0 1 0 0 dec

b 1 0 0 1 1 0 0 0 0 0 1 1 1 0 1 1 0 dec

Output pins

out 0 0 0 1 0 0 0 0 0 0 0 1 1 0 1 0 0 dec

Test Script Compare File Output File Diff Table

17 eval,

18 output;

19

20 set a %011111111111111111,

21 set b %011111111111111111,

22 eval,

23 output;

24

25 set a %01010101010101010,

26 set b %00101010101010101,

27 eval,

28 output;

29

30 set a %00011110011000011,

31 set b %00000111111110000,

32 eval,

33 output;

34

35 set a %00001001000110100,

36 set b %01001100001110110,

37 eval,

38 output;

Simulation successful: The output file is identical to the compare file

31°C Sunny

Search

10:37 AM 23-09-26

Software | nand2tetris

NAND2Tetris

ALU Code Request

sg562656-sudo/Nand2Tetris

Ask Gemini

nand2tetris.github.io/web-ide/chip/

Begin browsing instantly: Chrome can now launch when Windows starts. Allow

NAND2Tetris / Hardware Simulator / Or16.hdl

HDL

--LBSt

Project 1

Or16

Chip Or16

Eval Reset Clock: 0

Test

Steps: 9 Outputs: 7

5 /**

10 CHIP Or16 {

11 IN a[16], b[16];

12 OUT out[16];

13

14 PARTS:

15 //// Replace this comment with your code.

16 Or(a=a[0], b=b[0], out=out[0]);

17 Or(a=a[1], b=b[1], out=out[1]);

18 Or(a=a[2], b=b[2], out=out[2]);

19 Or(a=a[3], b=b[3], out=out[3]);

20 Or(a=a[4], b=b[4], out=out[4]);

21 Or(a=a[5], b=b[5], out=out[5]);

22 Or(a=a[6], b=b[6], out=out[6]);

23 Or(a=a[7], b=b[7], out=out[7]);

24 Or(a=a[8], b=b[8], out=out[8]);

25 Or(a=a[9], b=b[9], out=out[9]);

26 Or(a=a[10], b=b[10], out=out[10]);

27 Or(a=a[11], b=b[11], out=out[11]);

28 Or(a=a[12], b=b[12], out=out[12]);

29 Or(a=a[13], b=b[13], out=out[13]);

30 Or(a=a[14], b=b[14], out=out[14]);

31 Or(a=a[15], b=b[15], out=out[15]);

32

33 }

Input pins

a 0 0 0 1 0 0 1 0 0 0 1 1 0 1 0 0 dec

b 1 0 0 1 1 0 0 0 0 0 1 1 1 0 1 1 0 dec

Output pins

out 1 0 0 1 1 0 1 0 0 1 1 1 0 1 1 0 dec

Test Script

Compare File

Output File

Diff Table

17 eval,

18 output;

19

20 set a %011111111111111111,

21 set b %011111111111111111,

22 eval,

23 output;

24

25 set a %01010101010101010,

26 set b %00101010101010101,

27 eval,

28 output;

29

30 set a %00011110011000011,

31 set b %00000111111110000,

32 eval,

33 output;

34

35 set a %00001001000110100,

36 set b %01001100001110110,

37 eval,

38 output;

Simulation successful: The output file is identical to the compare file

31°C Sunny

Search

10:37 AM 23-09-26

Software | nand2tetris

NAND2Tetris

ALU Code Request

sg562656-sudo/Nand2Tetris

Ask Gemini

nand2tetris.github.io/web-ide/chip/

Begin browsing instantly: Chrome can now launch when Windows starts. Allow

NAND2Tetris / Hardware Simulator / Mux16.hdl

Chip Mux16 Eval Reset Clock: 0

Test Steps: 11 Outputs: 9

HDL --LBSt Project 1 Mux16

```
5 /**
9 */
10 CHIP Mux16 {
11     IN a[16], b[16], sel;
12     OUT out[16];
13
14     PARTS:
15         //// Replace this comment with your code.
16         Mux(a=a[0], b=b[0], sel=sel, out=out[0]);
17         Mux(a=a[1], b=b[1], sel=sel, out=out[1]);
18         Mux(a=a[2], b=b[2], sel=sel, out=out[2]);
19         Mux(a=a[3], b=b[3], sel=sel, out=out[3]);
20         Mux(a=a[4], b=b[4], sel=sel, out=out[4]);
21         Mux(a=a[5], b=b[5], sel=sel, out=out[5]);
22         Mux(a=a[6], b=b[6], sel=sel, out=out[6]);
23         Mux(a=a[7], b=b[7], sel=sel, out=out[7]);
24         Mux(a=a[8], b=b[8], sel=sel, out=out[8]);
25         Mux(a=a[9], b=b[9], sel=sel, out=out[9]);
26         Mux(a=a[10], b=b[10], sel=sel, out=out[10]);
27         Mux(a=a[11], b=b[11], sel=sel, out=out[11]);
28         Mux(a=a[12], b=b[12], sel=sel, out=out[12]);
29         Mux(a=a[13], b=b[13], sel=sel, out=out[13]);
30         Mux(a=a[14], b=b[14], sel=sel, out=out[14]);
31         Mux(a=a[15], b=b[15], sel=sel, out=out[15]);
32 }
```

Input pins

a 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 dec

b 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec

sel 1

Output pins

out 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec

Test Script Compare File Output File Diff Table

27 eval,

28 output;

29

30 set a %B1001100001110110,

31 set b %B0000000000000000,

32 set sel 0,

33 eval,

34 output;

35

36 set sel 1,

37 eval,

38 output;

39

40 set a %B1010101010101010,

41 set b %B0101010101010101,

42 set sel 0,

43 eval,

44 output;

45

46 set sel 1,

47 eval,

48 output;

Simulation successful: The output file is identical to the compare file

31°C Sunny

Search

10:37 AM 23-09-26

Software | nand2tetr...

NAND2Tetr...

ALU Code Request

sg562656-sudo/Nand2Tetr...

Ask Gemini

nand2tetr...

Begin browsing instantly: Chrome can now launch when Windows starts.

Allow

NAND2Tetr...

Hardware Simulator / Or8Way.hdl

Chip Or8Way

Eval Reset Clock: 0

Test

Steps: 8 Outputs: 6

HDL

--LBSt

Project 1

Or8Way

1 // This file is part of www.nand2tetr...

2 // and the book "The Elements of Computing Systems"

3 // by Nisan and Schocken, MIT Press.

4 // File name: projects/1/Or8Way.hdl

5 /**

6 * 8-way Or gate:

7 * out = in[0] Or in[1] Or ... Or in[7]

8 */

9 CHIP Or8Way {

10 IN in[8];

11 OUT out;

12

13 PARTS:

14 //// Replace this comment with your code.

15 Or(a=in[0], b=in[1], out=or01);

16 Or(a=in[2], b=in[3], out=or23);

17 Or(a=in[4], b=in[5], out=or45);

18 Or(a=in[6], b=in[7], out=or67);

19

20 Or(a=or01, b=or23, out=or0123);

21 Or(a=or45, b=or67, out=or4567);

22

23 Or(a=or0123, b=or4567, out=out);

24 }

Chip Or8Way

Eval Reset Clock: 0

Input pins

in 0 0 1 0 0 1 1 0 dec

Output pins

out 1

Internal pins

or01 1

or23 1

or45 1

or67 0

or0123 1

or4567 1

Test

Steps: 8 Outputs: 6

Slow Run Fast

Test Script Compare File Output File Diff Table

7 compare-to Or8Way.cmp,

8 output-list in%82.8.2 out;

9

10

11

12

13

14 set in %B11111111,

15 eval,

16 output;

17

18 set in %B00010000,

19 eval,

20 output;

21

22 set in %B00000001,

23 eval,

24 output;

25

26 set in %B00100110,

27 eval,

28 output;

Simulation successful: The output file is identical to the compare file

31°C Sunny

Search

10:38 AM 23-09-26

Begin browsing instantly: Chrome can now launch when Windows starts. Allow

NAND2Tetris / Hardware Simulator / Mux4Way16.hdl

HDL --LBSt Project 1 Mux4Way

```
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/Mux4Way16.hdl
5 /**
6  * 4-way 16-bit multiplexor:
7  * out = a if sel = 00
8  *   b if sel = 01
9  *   c if sel = 10
10 *   d if sel = 11
11 */
12 CHIP Mux4Way16 {
13     IN a[16], b[16], c[16], d[16], sel[2];
14     OUT out[16];
15
16     PARTS:
17         //// Replace this comment with your code.
18         Mux16(a=a, b=b, sel=sel[0], out=ab);
19         Mux16(a=c, b=d, sel=sel[0], out=cd);
20         Mux16(a=ab, b=cd, sel=sel[1], out=out);
21 }
```

Chip Mux4Way16 Eval Reset Clock: 0

Input pins

a 0 0 0 1 0 0 1 0 0 0 1 1 0 1 0 0 dec

b 1 0 0 1 1 0 0 0 0 1 1 1 0 1 1 0 dec

c 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 dec

d 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec

sel 1 1 dec

Output pins

out 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec

Internal pins

ab 1 0 0 1 1 0 0 0 0 1 1 1 0 1 1 0 dec

cd 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec

Test Steps: 11 Outputs: 9

Slow Fast

Test Script Compare File Output File Diff Table

```
20 set sel 0,
27 eval,
28 output;
29
30 set a %00001001000110100,
31 set b %01001100001110110,
32 set c %01010101010101010,
33 set d %00101010101010101,
34 set sel 0,
35 eval,
36 output;
37
38 set sel 1,
39 eval,
40 output;
41
42 set sel 2,
43 eval,
44 output;
45
46 set sel 3,
47 eval,
48 output;
```

Simulation successful: The output file is identical to the compare file

Software | nand2tetris

NAND2Tetris

ALU Code Request

sg562656-sudo/Nand2Tetris

Ask Gemini

nand2tetris.github.io/web-ide/chip/

Begin browsing instantly: Chrome can now launch when Windows starts. Allow

NAND2Tetris / Hardware Simulator / Mux8Way16.hdl

Chip Mux8Way16 Eval Reset Clock: 0

Test Steps: 19 Outputs: 17

HDL

--LBSt

Project 1

Mux8Way

```
5 /**
10 *      d if sel = 011
11 *      e if sel = 100
12 *      f if sel = 101
13 *      g if sel = 110
14 *      h if sel = 111
15 */
16 CHIP Mux8Way16 {
17     IN a[16], b[16], c[16], d[16],
18         e[16], f[16], g[16], h[16],
19         sel[3];
20     OUT out[16];
21
22     PARTS:
23         //// Replace this comment with your code.
24         Mux16(a=a, b=b, sel=sel[0], out=ab);
25         Mux16(a=c, b=d, sel=sel[0], out=cd);
26         Mux16(a=e, b=f, sel=sel[0], out=ef);
27         Mux16(a=g, b=h, sel=sel[0], out=gh);
28
29         Mux16(a=ab, b=cd, sel=sel[1], out=abcd);
30         Mux16(a=ef, b=gh, sel=sel[1], out=efgh);
31
32         Mux16(a=abcd, b=efgh, sel=sel[2], out=out);
33 }
```

Input pins

a

0 0 0 1 0 0 1 0 0 0 1 1 0 1 0 0

dec

b

0 0 1 0 0 0 1 1 0 1 0 0 0 1 0 1

dec

c

0 0 1 1 0 1 0 0 0 1 0 1 0 1 1 0

dec

d

0 1 0 0 0 1 0 1 0 1 1 0 0 1 1 1

dec

e

0 1 0 1 0 1 1 0 0 1 1 1 1 0 0 0

dec

f

0 1 1 0 0 1 1 1 1 0 0 0 1 0 0 1

dec

g

0 1 1 1 1 0 0 0 1 0 0 1 1 0 1 0

dec

h

1 0 0 0 1 0 0 1 1 0 1 0 1 0 1 1

dec

sel

1 1 1

dec

Output pins

out

1 0 0 0 1 0 0 1 1 0 1 0 1 0 1 1

dec

Internal pins

ab

0 0 1 0 0 0 1 1 0 1 0 0 0 1 0 1

dec

Test Script

Compare File

Output File

Diff Table

```
67 eval,
68 output;
69
70 set sel 3,
71 eval,
72 output;
73
74 set sel 4,
75 eval,
76 output;
77
78 set sel 5,
79 eval,
80 output;
81
82 set sel 6,
83 eval,
84 output;
85
86 set sel 7,
87 eval,
88 output;
```

Simulation successful: The output file is identical to the compare file

31°C Sunny

Search

ENG IN

69%

10:38 AM 23-09-26

Software | nand2tetris

NAND2Tetris

ALU Code Request

sg562656-sudo/Nand2Tetris

Ask Gemini

←

→

↺

🌐

nand2tetris.github.io/web-ide/chip/

🔍

🔖

☆

📁

📄

🔗

ⓘ

🌐

Begin browsing instantly: Chrome can now launch when Windows starts.

Allow

✕

NAND2Tetris / Hardware Simulator / DMux8Way.hdl

🔧

📅

📄

🖨

↔

🔍

🔖

📄

⚙

🔔

HDL

●

--LBSt

Project 1

DMux8Way

Chip DMux8Way

Eval

Reset

Clock: 0

5 /**

8 *

9 *

10 *

11 *

12 *

13 *

14 *

15 */

16 CHIP DMux8Way {

17 IN in, sel[3];

18 OUT a, b, c, d, e, f, g, h;

19

20 PARTS:

21 //// Replace this comment with your code.

22 DMux(in=in, sel=sel[2], a=abcd, b=efgh);

23

24 DMux(in=abcd, sel=sel[1], a=ab, b=cd);

25 DMux(in=efgh, sel=sel[1], a=ef, b=gh);

26

27 DMux(in=ab, sel=sel[0], a=a, b=b);

28 DMux(in=cd, sel=sel[0], a=c, b=d);

29 DMux(in=ef, sel=sel[0], a=e, b=f);

30 DMux(in=gh, sel=sel[0], a=g, b=h);

31 }

Input pins

in 1

sel 1 1 1 dec

Output pins

a 0

b 0

c 0

d 0

e 0

f 0

g 0

h 1

Internal pins

abcd 0

efgh 1

Test

Steps: 19

Outputs: 17

Slow

Fast

Test Script

Compare File

Output File

Diff Table

53 eval,

54 output;

55

56 set sel %B011,

57 eval,

58 output;

59

60 set sel %B100,

61 eval,

62 output;

63

64 set sel %B101,

65 eval,

66 output;

67

68 set sel %B110,

69 eval,

70 output;

71

72 set sel %B111,

73 eval,

74 output;

Simulation successful: The output file is identical to the compare file

31°C

🖱

🔍 Search

🌐

📅

📄

🖨

🔍

🔖

📄

⚙

🔔

ENG IN

🔊

🔋 69%

10:38 AM

23-09-26

carry 1

10:24 AM
23-09-26

Software | nand2tetris

NAND2Tetris

ChatGPT

Ask Gemini

nand2tetris.github.io/web-ide/chip/

Begin browsing instantly: Chrome can now launch when Windows starts. Allow

NAND2Tetris / Hardware Simulator / FullAdder.hdl

HDL

--LBSt

Project 2

FullAdder

Chip FullAdder

Eval

Reset

Clock: 0

Test

Steps: 7

Outputs: 0

1 // This file is part of www.nand2tetris.org

2 // and the book "The Elements of Computing Systems"

3 // by Nisan and Schocken, MIT Press.

4 // File name: projects/2/FullAdder.hdl

5 /**

6 * Computes the sum of three bits.

7 */

8 CHIP FullAdder {

9 IN a, b, c; // 1-bit inputs

10 OUT sum, // Right bit of a + b + c

11 carry; // Left bit of a + b + c

12

13 PARTS:

14 //// Replace this comment with your code.

15 HalfAdder(a=a, b=b, sum=sum1, carry=carry1);

16 HalfAdder(a=sum1, b=c, sum=sum, carry=carry2);

17 Or(a=carry1, b=carry2, out=carry);

18

19 }

Input pins

a 0

b 0

c 0

Output pins

sum 0

carry 0

Internal pins

sum1 0

carry1 0

carry2 0

Test Script

Compare File

Output File

Diff Table

1 // This file is part of www.nand2tetris.org

2 // and the book "The Elements of Computing Systems"

3 // by Nisan and Schocken, MIT Press.

4 // File name: projects/2/FullAdder.tst

5

6 load FullAdder.hdl,

7 compare-to FullAdder.cmp,

8 output-list a b c sum carry%B2.1.2;

9

10 set a 0,

11 set b 0,

12 set c 0,

13 eval,

14 output;

15

16 set c 1,

17 eval,

18 output;

19

20 set b 1,

21 set c 0,

22 eval,

23 output;

HDL code: No syntax errors

31°C Sunny

Search

ENG IN

75%

10:25 AM 23-09-26

Software | nand2tetris

NAND2Tetris

ALU Code Request

Ask Gemini

nand2tetris.github.io/web-ide/chip/

Begin browsing instantly: Chrome can now launch when Windows starts. Allow

NAND2Tetris / Hardware Simulator / Add16.hdl

Eval Reset Clock: 0

Test Steps: 9 Outputs: 7

HDL

--LBSt

Project 2

Add16

5 /**

8 */

9 CHIP Add16 {

10 IN a[16], b[16];

11 OUT out[16];

12

13 PARTS:

14 HalfAdder(a=a[0], b=b[0], sum=out[0], carry=c1);

15

16 FullAdder(a=a[1], b=b[1], c=c1, sum=out[1], carry

17 FullAdder(a=a[2], b=b[2], c=c2, sum=out[2], carry

18 FullAdder(a=a[3], b=b[3], c=c3, sum=out[3], carry

19 FullAdder(a=a[4], b=b[4], c=c4, sum=out[4], carry

20 FullAdder(a=a[5], b=b[5], c=c5, sum=out[5], carry

21 FullAdder(a=a[6], b=b[6], c=c6, sum=out[6], carry

22 FullAdder(a=a[7], b=b[7], c=c7, sum=out[7], carry

23 FullAdder(a=a[8], b=b[8], c=c8, sum=out[8], carry

24 FullAdder(a=a[9], b=b[9], c=c9, sum=out[9], carry

25 FullAdder(a=a[10], b=b[10], c=c10, sum=out[10], c

26 FullAdder(a=a[11], b=b[11], c=c11, sum=out[11], c

27 FullAdder(a=a[12], b=b[12], c=c12, sum=out[12], c

28 FullAdder(a=a[13], b=b[13], c=c13, sum=out[13], c

29 FullAdder(a=a[14], b=b[14], c=c14, sum=out[14], c

30 FullAdder(a=a[15], b=b[15], c=c15, sum=out[15], c

31 }

Chip Add16

Input pins

a 0 0 0 1 0 0 1 0 0 0 1 1 0 1 0 0 dec

b 1 0 0 1 1 0 0 0 0 0 1 1 1 0 1 1 0 dec

Output pins

out 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 dec

Internal pins

c1 0

c2 0

c3 1

c4 0

c5 1

c6 1

c7 1

c8 0

c9 0

Test

Test Script

Compare File

Output File

Diff Table

17 eval,

18 output;

19

20 set a %011111111111111111,

21 set b %011111111111111111,

22 eval,

23 output;

24

25 set a %01010101010101010,

26 set b %00101010101010101,

27 eval,

28 output;

29

30 set a %00011110011000011,

31 set b %00000111111110000,

32 eval,

33 output;

34

35 set a %00001001000110100,

36 set b %01001100001110110,

37 eval,

38 output;

Simulation successful: The output file is identical to the compare file

31°C Sunny

Search

10:27 AM 23-09-26

Software | nand2tetris

NAND2Tetris

ALU Code Request

Ask Gemini

← → ↺

nand2tetris.github.io/web-ide/chip/

🔍 📄 ⭐ 🗑️ | S

Begin browsing instantly: Chrome can now launch when Windows starts.

Allow

✕

NAND2Tetris / Hardware Simulator / Inc16.hdl

<>

HDL --LBSt Project 2 ▾ Inc16 ▾

```
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/2/Inc16.hdl
5 /**
6  * 16-bit incrementer:
7  * out = in + 1
8  */
9 CHIP Inc16 {
10     IN in[16];
11     OUT out[16];
12
13     PARTS:
14     Add16(a=in, b[0]=true, b[1..15]=false, out=out);
15 }
```

Chip Inc16

Eval Reset Clock: 0

Input pins

in 1 1 1 1 1 1 1 1 1 1 1 1 1 0 1 1

Output pins

out 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0

Test Steps: 7 Outputs: 5

Slow Fast

Test Script

Compare File

Output File

Diff Table

```
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/2/Inc16.tst
5
6 load Inc16.hdl,
7 compare-to Inc16.cmp,
8 output-list in%B1.16.1 out%B1.16.1;
9
10 set in %0000000000000000, // in = 0
11 eval,
12 output;
13
14 set in %0111111111111111, // in = -1
15 eval,
16 output;
17
18 set in %00000000000000101, // in = 5
19 eval,
20 output;
21
22 set in %01111111111111011, // in = -5
23 eval,
24 output;
```

Simulation successful: The output file is identical to the compare file

31°C Sunny

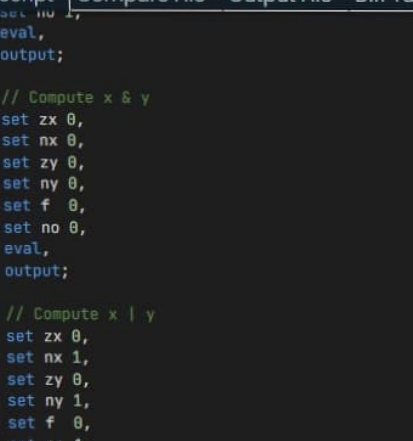
Search

ENG IN

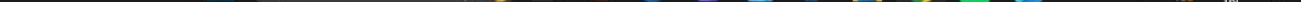
10:28 AM 23-09-26

X

Test Steps: 41 Outputs: 37

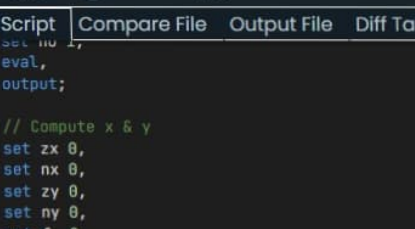


```
Test Script | Compare File | Output File | Diff Table
352 set no 1,
353 eval,
354 output;
355
356 // Compute x & y
357 set zx 0,
358 set nx 0,
359 set zy 0,
360 set ny 0,
361 set f 0,
362 set no 0,
363 eval,
364 output;
365
366 // Compute x | y
367 set zx 0,
368 set nx 1,
369 set zy 0,
370 set ny 1,
371 set f 0,
372 set no 1,
373 eval,
374 output;
```



Begin browsing instantly: Chrome can now launch when Windows starts.

Allow



The screenshot shows the AWS CloudShell console with the 'Test Script' editor open. The top bar indicates 'Steps: 41 Outputs: 37'. Below the tabs, a table lists the outputs. The script content is as follows:

```

352 set no 1,
353 eval,
354 output;
355
356 // Compute x & y
357 set zx 0,
358 set nx 0,
359 set zy 0,
360 set ny 0,
361 set f 0,
362 set no 0,
363 eval,
364 output;
365
366 // Compute x | y
367 set zx 0,
368 set nx 1,
369 set zy 0,
370 set ny 1,
371 set f 0,
372 set no 1,
373 eval,
374 output;

```

32°C Sunny Search [Taskbar icons: Edge, Word, PowerPoint, Teams, OneDrive, File Explorer, Chrome, WhatsApp, Photos] ENG IN 10:20 AM 23-09-2024 78%